Reduced-Sector χ-Universality in Lorentzian EC+NY Minisuperspace: Topology-Robust Admissibility, P-Channel Diagnostics, and Reduced Vacuum Orbits

Muacca

Version: v1
First release: 2026-05-16

Abstract

We analyse the topology dependence of the Lorentzian Einstein–Cartan plus Nieh–Yan (EC+NY) reduced sector across the four homogeneous three-geometries S^3, T^3, Nil^3, and Sol^3, restricted to the $\alpha = 0$ EC+NY reduced branch. For Nil^3 and Sol^3 the treatment is confined to the isotropic-scale reduction.

In this reduced sector the Hamiltonian constraint and the torsion auxiliary equations exhibit a topology-robust admissibility pattern: the EH, AX, and VT branches are admissible, while the MX branch is conditionally admissible through a real auxiliary branch. For Pontryagin-type quantities we separate the form-Hodge density P_{form}, the internal-pair diagnostic P_{int}, and the Nieh–Yan endpoint channel Q_{NY}. The form-Hodge density P_{form} vanishes exactly on the active torsion branches by block orthogonality. The diagnostic P_{int} on the MX branch carries the topology dependence in a single coefficient, $C_{\mathrm{topology}} = -9\chi$.

Furthermore, the reduced vacuum orbit atlas on the auxiliary shell reduces to $\dot{q}^2 + \chi = 0$.

Consequently, $\chi > 0$ admits no real initial data on the reduced vacuum atlas, $\chi = 0$ gives static or degenerate behaviour, and $\chi < 0$ gives monotonic expansion together with singular-approach collapse sheets.

The two relations $C_{\mathrm{topology}} = -9\chi$ and $\dot{q}^2 + \chi = 0$ are obtained from independent computational routes and converge on the same scalar χ; this convergence is the content of the χ-controlled topology universality. The results are a classification within the reduced Lorentzian EC+NY sector; global, anisotropic, and matter-coupled extensions are left to future work.

1 Introduction

1.1 Background and motivation

The DPPU series has progressively organised the Einstein–Cartan plus Nieh–Yan setting [7, 8] in terms of homogeneous geometries, torsion modes, Pontryagin-type diagnostics, and topology response [1, 2, 3, 4, 5]. Paper V [6] constructed the $\alpha = 0$ EC+NY baseline in Lorentzian signature on the S^3 closed-FLRW branch, and established the channel separation among the Hamiltonian constraint, the torsion auxiliary sector, and the diagnostic quantities P_{form}, P_{int}, and Q_{NY}.

The aim of the present paper is to extend that baseline to the four topologies

$$S^3, \quad T^3, \quad \mathrm{Nil}^3, \quad \mathrm{Sol}^3, \tag{1}$$

and to make explicit how topology dependence appears in the branch classification and in the diagnostic structure. The subject of this paper is not bounce cosmology model building, but a classificatory analysis of χ-controlled topology universality in the reduced sector.

1.2 Main results

The main results of this paper are the following.

1. In the Lorentzian EC+NY reduced setting, the four topologies S^3, T^3, Nil^3, and Sol^3 share the same local admissibility pattern: EH, AX, and VT are admissible, while MX is conditionally admissible.
2. On the active torsion branches, P_{form} vanishes exactly by block orthogonality.
3. $P_{\mathrm{int}}^{\mathrm{MX}}$ has a topology-universal functional template; the topology dependence is localised in the single coefficient $C_{\mathrm{topology}} = -9\chi$.
4. Q_{NY} is an endpoint or boundary channel: zero on VT, on-shell trivial on AX, and boundary-conditional on MX.
5. On the auxiliary shell, the reduced vacuum orbit is governed by $\dot{q}^2 + \chi = 0$.
6. Within this vacuum reduced atlas, no bounce-like or recollapse-like orbit class is found.

The scalar χ in result 3 and in result 5 is obtained from independent routes – a symbolic analysis of $P_{\mathrm{int}}^{\mathrm{MX}}$ and a derivation from the EH-spatial-curvature sector, respectively – and the agreement of these two routes is the substance of χ-universality.

In other words, in the Lorentzian EC+NY reduced sector topology does not alter the algebraic admissibility pattern, but controls the MX internal diagnostic and the vacuum auxiliary-shell orbit through the same scalar χ.

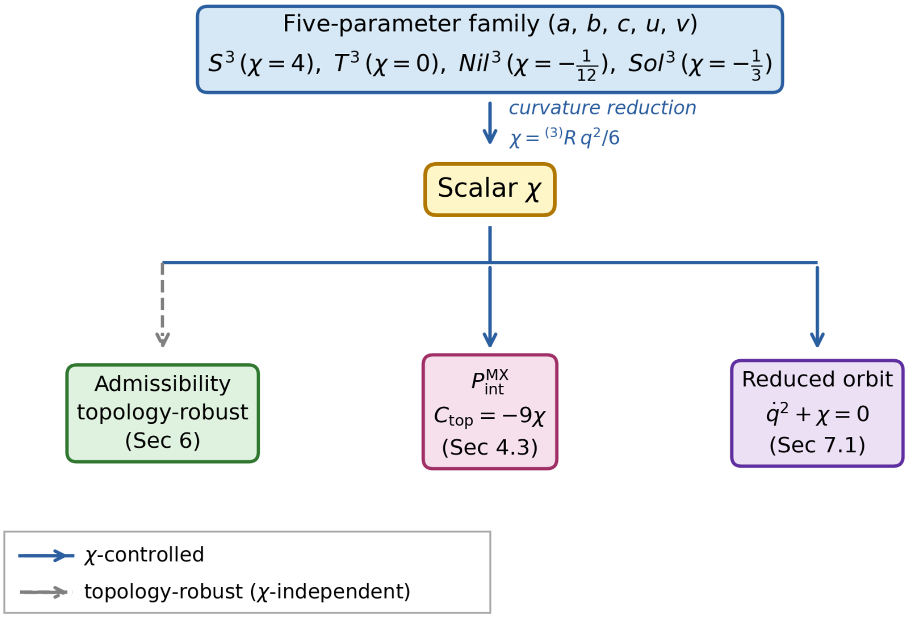

Figure 1: Three-level architecture of χ-controlled topology universality in the reduced LZ-native isotropic EC+NY sector. The five-parameter family (a, b, c, u, v) at the top reduces, via the spatial-curvature contraction, to the scalar χ, from which three independent consequences follow: the topology-robust admissibility pattern, the MX diagnostic identity $C_{\mathrm{topology}} = -9\chi$, and the reduced vacuum orbit $\dot{q}^2 + \chi = 0$. The arrow to admissibility (dashed) indicates that the admissibility pattern does not depend on χ. The arrows to $P^{\mathrm{MX}}_{\mathrm{int}}$ and to the reduced orbit (solid) indicate χ-mediated control. *Scope*: limited to the reduced LZ-native isotropic EC+NY sector; global, anisotropic, and matter-coupled extensions lie outside the scope of this paper.

1.3 Relation to earlier DPPU papers

Paper V [6] constructed the Lorentzian $\alpha = 0$ EC+NY baseline on S^3 and separated P_{form} from P_{int}. The present paper extends that baseline to a four-topology table.

In particular, the S^3 row of the present paper reproduces the $\alpha = 0$ Lorentzian baseline of paper V. In Table B1, the S^3 entry has $F_{\mathrm{shell}} = 4$, the MX auxiliary Hessian $-4q^2(\kappa^4\theta^2_{\mathrm{NY}} + 1)/9$, and the classification label `L_CONDITIONALLY_ADMISSIBLE`. The new content of the present paper is that the same pattern extends to T^3, Nil^3, and Sol^3.

In paper IV [5] the same four Thurston geometries [9, 10] produced four response types

– none, uniaxial, triaxial, and biaxial – in the Euclidean response dictionary. As discussed in Section 5.3, the same four geometries reappear here as structurally distinguished loci in the LZ-native isotropic five-parameter family. This correspondence is a structural resonance, and its interpretation is taken up in Section 8.2.

1.4 Organisation of the paper

Section 2 fixes the Lorentzian-native setup and the topology conventions. Section 3 states the Hamiltonian admissibility backbone. Section 4 organises the channel dictionary for P_{form}, P_{int}, and Q_{NY}. Section 5 connects the diagnostic-level and reduced-orbit-level χ-universality. Section 6 summarises the four-topology admissibility classification. Section 7 states the reduced orbit atlas and the χ-sign separation. Section 8 discusses the meaning of χ-universality, the relation to paper IV, and future directions. Appendix A supplements the topology conventions, local reductions, and caveats. Appendix B collects tables, computational checks, and reproducibility notes. Appendix C states a Bianchi I lite comparison and the reduced-atlas scope guard.

2 Lorentzian-native setup and topology conventions

2.1 Reduced Lorentzian frame

We use the orthonormal frame of Lorentzian signature $(-,+,+,+)$. The coframe is written as

$$e^0 = N(t)\,dt, \qquad e^i = q(t)\,\sigma^i_\Sigma, \tag{2}$$

where $N(t)$ is the lapse, $q(t)$ is the isotropic scale common to all topologies Σ, and σ^i_Σ are the left-invariant coframes on Σ. Torsion variables are taken to be the axial component $\eta(t)$ and the vector-trace component $V(t)$, giving the pure branches AX and VT and the mixed branch MX.

The normalisation of the spatial 3-curvature is fixed by

$$^{(3)}R = \frac{6\chi}{q^2}, \tag{3}$$

and χ is used as the topology scalar throughout this paper. The scalar χ is derived from the EH (pure Einstein–Hilbert) sector only and does not depend on the torsion variables.

2.2 Topology structure constants and χ values

The structure constants of the four topologies are specialisations of the five-parameter LZ-native isotropic family, following the class-A / class-B classification of Bianchi-type homogeneous spaces [11] and the three-topology mode dictionary developed earlier in the DPPU series [4]:

$$C^1{}_{23} = \frac{a}{q}, \quad C^2{}_{31} = \frac{b}{q}, \quad C^3{}_{12} = \frac{c}{q}, \quad C^2{}_{12} = \frac{u}{q}, \quad C^3{}_{13} = \frac{v}{q}, \tag{4}$$

where the subscripts are antisymmetric. Then

$$^{(3)}R\,q^2 = -\tfrac{1}{2}\Big(a^2 - 2ab - 2ac + b^2 - 2bc + c^2 + 4u^2 + 4uv + 4v^2\Big) =: -\tfrac{1}{2}\,\Delta, \tag{5}$$

and comparison with the defining relation $^{(3)}R = 6\chi/q^2$ gives

$$6\chi = -\frac{\Delta}{2}, \qquad \chi = -\frac{\Delta}{12}. \tag{6}$$

The specialised values for each topology are listed in Table 1.

Topology	(a,b,c,u,v)	χ	Reduction status	Caveat
S^3	$(4,4,4,0,0)$	4	closed isotropic reference	—
T^3	$(0,0,0,0,0)$	0	flat reference	—
Nil^3	$(0,0,-1,0,0)$	$-1/12$	isotropic-scale local branch	anisotropic EOM left to future work
Sol^3	$(0,0,0,1,-1)$	$-1/3$	isotropic-scale local branch	compact quotient outside the scope of this paper

Table 1: Topology conventions, χ values, and caveats.

We treat all four geometries in a common reduced-topology table. For Nil^3 and Sol^3, the entries refer to the isotropic-scale reductions used in this paper; global quotient and anisotropic extensions are left to future work.

2.3 EC+NY reduced action

The Lorentzian EC+NY reduced action [7, 8] for each topology Σ and torsion mode is written in the form

$$S_\Sigma = \int dt\, A_\Sigma\, q\left(N\, F_\Sigma(q,\eta,V;\chi) - \frac{\dot{q}^2}{N}\right). \tag{7}$$

Here A_Σ is a spatial normalisation factor independent of χ. The effective potential F_Σ is determined by the topology and the torsion branch, and its explicit form is used in Section 3. The auxiliary variables $\eta(t)$ and $V(t)$ carry no velocity terms in the action and are treated algebraically as an auxiliary sector.

The Nieh–Yan term θ_{NY} [8] does not appear in the bulk F_Σ on AX and VT, and appears on MX only as a mixing term $\propto V\eta\,\kappa^2\theta_{\mathrm{NY}}$. In this paper the bulk dynamical analysis is taken as primary, and Q_{NY} is separated as an endpoint / boundary channel (Section 4.4).

3 Hamiltonian admissibility by topology

3.1 Admissibility criterion

In this paper, admissibility in the reduced sector is judged by the following three criteria.

1. *EH constraint closure*: the Hamiltonian constraint of the pure EH part closes under the Lorentzian convention (`EH_PASS`).

2. *Reduced action skeleton*: the EC+NY reduced Lagrangian of each torsion mode has a well-defined velocity structure (`LRED_PASS`).

3. *Torsion auxiliary closure*: the algebraic equations for the auxiliary variables η and V admit real solutions (`AUX_PASS` or `AUX_CONDITIONAL_REAL_BRANCH`).

This notion of admissibility is not a claim of orbit stability, unitarity, or full constraint closure.

3.2 Hamiltonian constraint

Lapse variation $\delta S/\delta N = 0$ gives the Hamiltonian constraint

$$\mathcal{H} \equiv A_\Sigma\, q\left(F_\Sigma + \frac{\dot{q}^2}{N^2}\right) = 0, \qquad \text{that is,} \quad F_\Sigma + \frac{\dot{q}^2}{N^2} = 0. \tag{8}$$

On the EH branch ($\eta = V = 0$) one has $F_\Sigma\big|_{\mathrm{EH}} = \chi$ (with the χ of Table 1 substituted for each topology), which is the content of `EH_PASS`:

Topology	F_{EH} (on-shell)	Constraint residual
S^3	4	$4N^2 + \dot{r}^2 = 0$
T^3	0	$\dot{r}^2 = 0$
Nil^3	$-1/12$	$N^2 - 12\dot{R}^2 = 0$
Sol^3	$-1/3$	$3N^2 - 9\dot{q}^2 = 0$ (local/iso)

Table 2: EH residuals for each topology.

The torsion auxiliary equations are the algebraic equations $\delta S/\delta\eta = 0$ and $\delta S/\delta V = 0$. On AX only η is non-trivial and on VT only V, each yielding `AUX_PASS`. On MX the auxiliary Hessian determinant is

$$\det\begin{pmatrix} \partial_\eta(\partial_\eta F) & \partial_V(\partial_\eta F) \\ \partial_\eta(\partial_V F) & \partial_V(\partial_V F) \end{pmatrix} = -\frac{4q^2(\kappa^4\theta_{\mathrm{NY}}^2 + 1)}{9} < 0. \tag{9}$$

For $q > 0$ in the reduced-sector domain and real κ and θ_{NY}, one has $\kappa^4\theta_{\mathrm{NY}}^2 + 1 > 0$, so the auxiliary Hessian is non-degenerate. We retain this mixed auxiliary channel as a branch with real-branch selection and origin tagging, and record it as `AUX_CONDITIONAL_REAL_BRANCH`, that is, as the `H_CONDITIONAL` status of MX.

The negative determinant does not imply a stable minimum. The notion of *conditional admissibility* used here is a weak criterion requiring only algebraic closure and real-branch selection at the non-degenerate mixed auxiliary saddle, and is independent of the Hessian signature (dynamical stability).

3.3 Mode status

Collecting the above criteria gives Table 3. The result is the same across the four topologies.

Mode	Hamiltonian status	Interpretation
EH	`H_PASS`	pure EH reference branch
AX	`H_PASS`	active-torsion admissible
VT	`H_PASS`	active-torsion admissible
MX	`H_CONDITIONAL`	conditional admissibility via real auxiliary branch

Table 3: Hamiltonian status by topology and mode.

The `H_CONDITIONAL` label for MX is not an obstruction or a failure. Assuming the existence of a real auxiliary branch, the off-shell behaviour of the subsequent P-channel diagnostics and the Q_{NY} boundary channel is recorded as a diagnostic, which is the content of conditional admissibility.

Remark. `H_PASS` and `L_ADMISSIBLE` do not imply dynamical stability. The Sol^3 table entries refer to the local/isotropic reduction and do not constitute a full resolution of the compact quotient.

4 P-channel dictionary and topology response

4.1 Channel separation

The Pontryagin-type quantities are separated into three channels:

Channel	Result	Role	Distinction
P_{form}	Exactly zero on active torsion branches	Structural cancellation	Distinct from dynamical / orbit stability
P_{int}	MX internal-pair diagnostic; $C_{\text{topology}} = -9\chi$	Diagnostic topology response	Not a true Pontryagin density
Q_{NY}	Endpoint / boundary channel: zero on VT, on-shell trivial on AX, boundary-conditional on MX	Boundary bookkeeping	Not a bulk dynamical mechanism

Table 4: P-channel dictionary.

P_{int} is a diagnostic quantity based on the contraction of internal index pairs in the orthonormal frame, and is *not* a form-Hodge Pontryagin density [13]. This distinction is essential for the interpretation given in this paper.

4.2 P_{form} cancellation by block orthogonality

P_{form} is defined as the form-Hodge contraction of the curvature 2-form. In components,

$$P_{\text{form}} = \frac{1}{4} \sum_{a,b,c,d} R^{ab\,cd} \, (*R_{ab})_{cd}, \qquad (*R_{ab})_{cd} = \frac{1}{2} \epsilon_{cd\,ef} \, R_{ab}{}^{ef}, \tag{10}$$

with the Hodge dual taken in the Lorentzian metric-aware form-index sense.

Lemma 1 (Block-orthogonality cancellation)**.** *In the Lorentzian isotropic EC+NY reduced sector studied here, for each active torsion branch* AX, VT, MX *and for each topology* S^3*,* T^3*,* Nil^3*,* Sol^3*, the form-Hodge Pontryagin density satisfies*

$$P_{\text{form}} = 0 \tag{11}$$

identically.

Proof sketch. Under the reduced coframe $e^0 = N\,dt$, $e^i = q\,\sigma^i$, the non-zero curvature 2-forms split, for each internal pair, into a lapse-space block and a purely spatial block. The form-Hodge dual maps each block to its complement, but no internal pair admits a contractible partner, so the pairwise contraction vanishes. Symbolic verification of all branches is recorded in Appendix B. This cancellation is a structural result of the isotropic reduced setting considered here, and does not hold in general in the full theory or under anisotropic extensions. Moreover, $P_{\text{form}} = 0$ is not a claim of dynamical stability. $\square$

4.3 $P_{\text{int}}^{\text{MX}}$ as internal-pair diagnostic

On AX and on VT, the off-shell expression for P_{int} vanishes identically. On MX an off-shell non-zero channel appears, but it returns to zero on the auxiliary shell ($\eta = V = 0$ under the real-branch condition). This off-shell behaviour does not violate the Hamiltonian admissibility.

On the MX branch, the off-shell $P_{\text{int}}^{\text{MX}}$ takes the common functional template across the four topologies,

$$P_{\text{int}}^{\text{MX}} = \frac{2V\eta(-V^2q^2 + 9\eta^2 + C_{\text{topology}})}{9q^3}, \tag{12}$$

and the topology dependence is localised entirely in the single coefficient C_{topology}:

Topology	χ	C_{topology}	$C_{\text{topology}} + 9\chi$
S^3	4	-36	0
T^3	0	0	0
Nil^3	$-1/12$	$3/4$	0
Sol^3	$-1/3$	3	0

Table 5: $C_{\text{topology}} = -9\chi$ bridge table.

The coefficient C_{topology} is extracted from the raw symbolic expression of $P_{\text{int}}^{\text{MX}}$ by template fitting, without reference to χ. The scalar χ is obtained independently from the EH / spatial-curvature sector (see Section 2.2). The agreement of these two quantities is the substance of the χ-bridge result discussed in Section 5.

4.4 Q_{NY} as endpoint channel

The Nieh–Yan primitive [8, 13] is $Q_{\text{NY}} = e^a \wedge T_a$, and we treat it as an exact-form endpoint quantity independent of P_{form} and P_{int}. The branches AX and MX carry an active axial torsion η, and the reduced pullback reads

$$Q_{\text{NY}} = \frac{6\eta}{q}\,e^{123}, \qquad \int_\Sigma Q_{\text{NY}} = 6\mathcal{V}_\Sigma\,q^2\eta, \tag{13}$$

with orientation fixed by $Q(t_f) - Q(t_i)$. Here $\mathcal{V}_\Sigma$ is the unit-coframe volume normalisation factor, which is distinguished from the torsion variable $V(t)$. On the VT branch, the EOM imposes $\eta = 0$, so $Q_{\rm NY} = 0$ already off-shell. On the auxiliary shell, AX is on-shell trivial and MX is boundary-conditional. Thus $Q_{\rm NY}$ is not a bulk dynamical mechanism but should be treated as endpoint / boundary bookkeeping.

5 χ-universality across diagnostic and reduced-orbit levels

5.1 Two independent appearances of χ

The central claim of this paper is the bridge result that two independent computations converge on the same scalar χ.

Level	Equation	Derivation route
Diagnostic	$C_{\text{topology}} = -9\chi$	template extraction from raw $P^{\text{MX}}_{\text{int}}$
Reduced orbit	$\dot{q}^2 + \chi = 0$	vacuum orbit atlas on the auxiliary shell

Table 6: χ-universality summary.

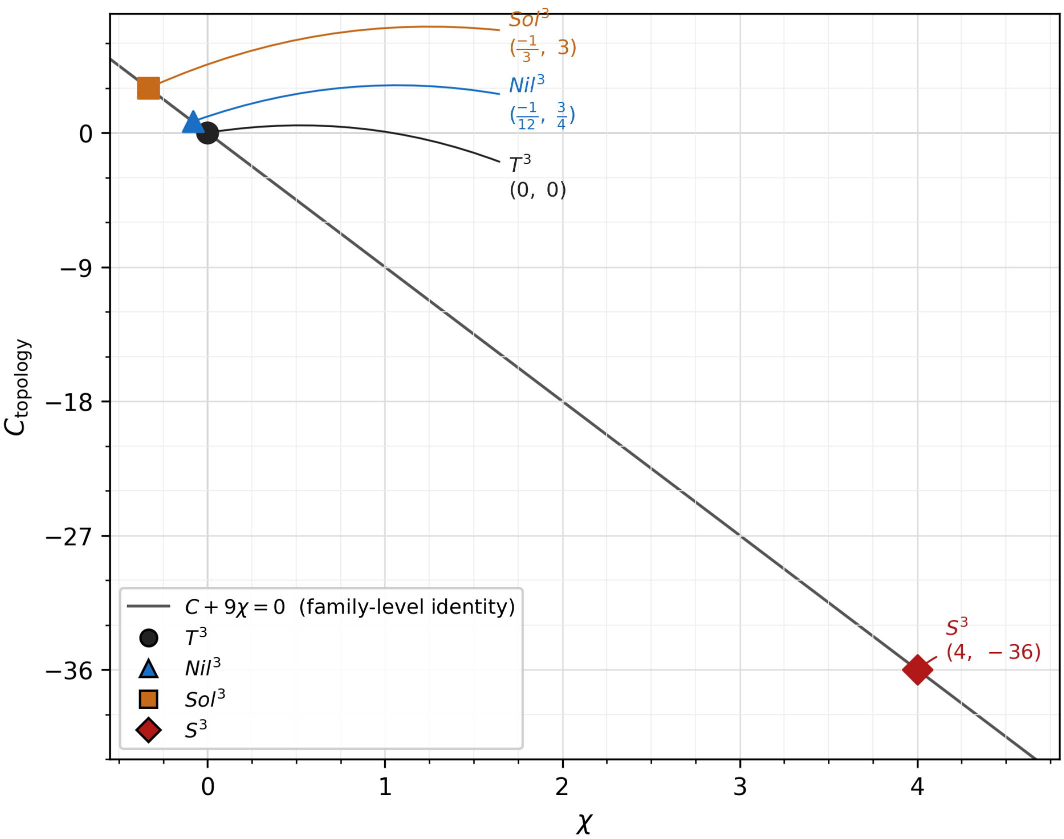

Figure 2: Diagnostic bridge: the $C_{\text{topology}} = -9\chi$ line. The horizontal axis is χ (derived independently from the LC spatial curvature) and the vertical axis is C_{topology} (extracted from the raw symbolic expression of $P^{\text{MX}}_{\text{int}}$ by template fitting, without reference to χ). The four points T^3, Nil^3, Sol^3, S^3 all lie exactly on the line $C_{\text{topology}} + 9\chi = 0$. The line is the graphical representation of the family-level identity of Section 5.2; the four points are its specialisations. *Scope*: a family-level identity within the LZ-native isotropic MX sector; $P^{\text{MX}}_{\text{int}}$ is an internal-pair diagnostic, not a true Pontryagin density.

The independence of these two quantities deserves emphasis. As stated in Section 4.3, C_{topology} is a coefficient extracted from the raw symbolic expression of $P_{\text{int}}^{\text{MX}}$ without reference to χ. In contrast, χ is derived independently from the spatial-curvature sector as

$$\chi = \frac{{}^{(3)}R\, q^2}{6}. \tag{14}$$

Neither computation takes the other quantity as an input, so the relation $C_{\text{topology}} = -9\chi$ is not a rephrasing of a definition but the bridge result between two independent computational routes.

5.2 Five-parameter family-level identity

Over the full five-parameter LZ-native isotropic family of Section 2.2, the family-level expression for C_{topology} is

$$C_{\text{topology}} = \frac{3}{4}\,\Delta = \frac{3}{4}\Big(a^2 - 2ab - 2ac + b^2 - 2bc + c^2 + 4u^2 + 4uv + 4v^2\Big)\,, \tag{15}$$

and since $\chi = -\Delta/12$,

$$C_{\text{topology}} + 9\chi = \frac{3\Delta}{4} - \frac{9\Delta}{12} = 0 \tag{16}$$

holds as an exact symbolic identity. This identity is a structural identity over the entire five-parameter family, not just at the four specific numerical values of the topologies; the algebraic core is given in Appendix B.

5.3 Distinguished topology loci

The four topologies are not arbitrary sample points of the five-parameter family but are located at structurally distinguished loci (Table 7).

Topology	(a, b, c, u, v)	χ	$C_{\text{topology}} = -9\chi$	Structural locus	paper IV echo
T^3	$(0, 0, 0, 0, 0)$	0	0	origin	none
Nil^3	$(0, 0, -1, 0, 0)$	$-1/12$	$3/4$	single class-A axis	uniaxial
S^3	$(4, 4, 4, 0, 0)$	4	-36	isotropic class-A diagonal	triaxial
Sol^3	$(0, 0, 0, 1, -1)$	$-1/3$	3	balanced class-B diagonal pair	biaxial

Table 7: Distinguished topology loci and paper IV echo.

The "paper IV echo" column records an interpretive correspondence with the KK Higgsing response dictionary of paper IV (none / uniaxial / triaxial / biaxial); details are given in Section 8.2.

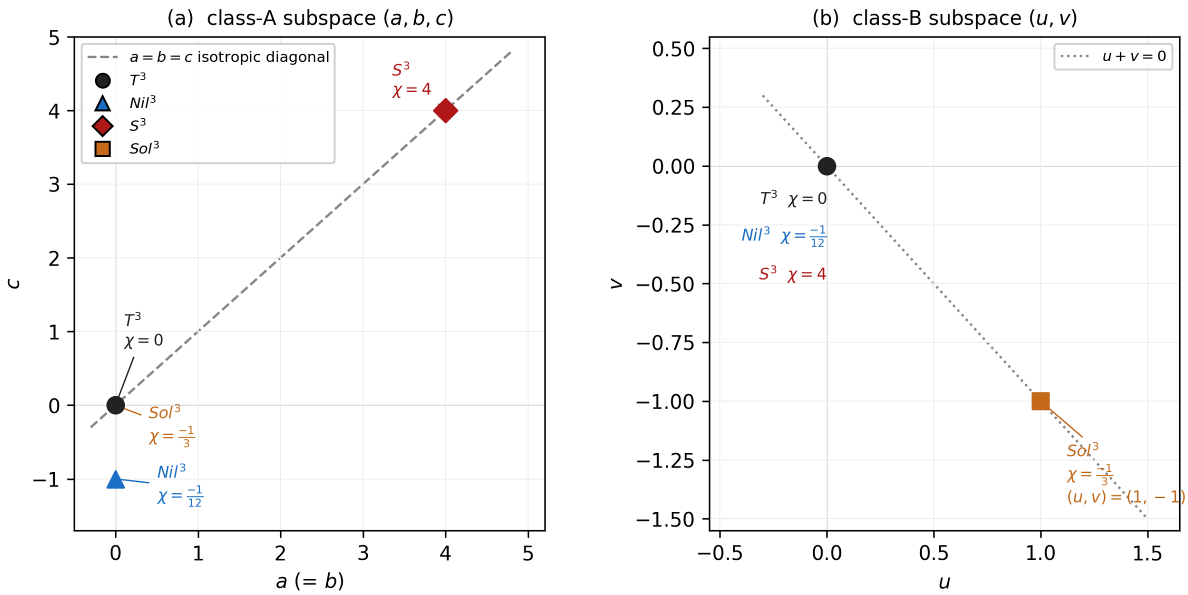

Figure 3: Distinguished loci in the five-parameter family (a, b, c, u, v). **(a) class-A subspace** (a, b, c): all topologies lie in the plane $b = a$ and are shown in the (a, c) coordinates. T^3 and Sol^3 sit at the class-A origin, Nil^3 on a single axis, and S^3 on the isotropic diagonal $a = b = c$ (dashed). **(b) class-B subspace** (u, v): T^3, Nil^3, and S^3 collapse to the $u = v = 0$ origin, while Sol^3 sits at the balanced pair $(u, v) = (1, -1)$ on the line $u+v = 0$ (dotted). The four points are structurally distinguished loci of the family, not arbitrary samples. *Caveat*: the "paper IV echo" label indicates a structural resonance with the paper IV KK Higgsing dictionary, not an identification of the underlying mechanism or theorem (see Section 8.2). The present family is the LZ-native isotropic MX family, not a full-theory universal family.

The scope of χ-universality is restricted to the LZ-native isotropic MX reduced sector of this paper. Extensions to full-theory universality, arbitrary anisotropic EOM, or matter-coupled sectors are not claimed here.

6 Four-topology admissibility classification

The local admissibility classification that integrates the Hamiltonian admissibility of Section 3 and the P-channel diagnostics of Section 4 is collected in Table 8.

Topology	EH	AX	VT	MX
S^3	L_ADMISSIBLE	L_ADMISSIBLE	L_ADMISSIBLE	L_CONDITIONALLY_ADMISSIBLE
T^3	L_ADMISSIBLE	L_ADMISSIBLE	L_ADMISSIBLE	L_CONDITIONALLY_ADMISSIBLE
Nil^3	L_ADMISSIBLE	L_ADMISSIBLE	L_ADMISSIBLE	L_CONDITIONALLY_ADMISSIBLE
Sol^3	L_ADMISSIBLE	L_ADMISSIBLE	L_ADMISSIBLE	L_CONDITIONALLY_ADMISSIBLE

Table 8: Four-topology admissibility classification.

The label `L_ADMISSIBLE` indicates that the Hamiltonian constraint closes (Section 3.2), the active-torsion auxiliary closes (`AUX_PASS`), and the diagnostic checks $P_{\text{form}} = 0$ and $P_{\text{int}} = 0$ are passed. EH is a pure-curvature reference to which the P-channels are not applied. The label `L_CONDITIONALLY_ADMISSIBLE` indicates that the MX branch becomes admissible only under the real-auxiliary-branch condition $\kappa^4\theta_{\text{NY}}^2 + 1 > 0$ and the auxiliary-shell interpretation $\eta = V = 0$. The off-shell $P_{\text{int}}^{\text{MX}}$ and Q_{NY} are recorded as diagnostic / boundary channels but do not obstruct admissibility.

The structural points of this table can be summarised in three statements. First, the EH / AX / VT / MX mode pattern agrees across the four topologies. Second, the conditions of the conditional admissibility of MX are also common to the four topologies (see the auxiliary determinant in Section 3.2), so topology does not change the presence or absence of these conditions. Third, although Sol^3 carries a caveat regarding the compact quotient (see Appendix A), in the main classification table its row structure coincides with that of the other three topologies.

This topology-robust pattern is the background for the χ-universality of Section 5. Since admissibility itself does not vary with topology, topology dependence appears not in the admissibility label but in the diagnostic / orbit quantities C_{topology} and χ. In this sense, the bridge result of Section 5 is a natural consequence that rides on top of the admissibility classification.

7 Reduced orbit atlas and χ-sign separation

7.1 Derivation of the auxiliary-shell orbit equation

Taking the lapse variation of the reduced action of Section 2.3 gives the Hamiltonian constraint

$$\dot{q}^2 + F_\Sigma(q, \eta, V; \chi) = 0 \qquad (N = 1 \text{ gauge}), \tag{17}$$

with the sign in agreement with Section 3.2. The auxiliary fields of each torsion branch are determined algebraically from $\delta S/\delta\eta = \delta S/\delta V = 0$. The branch functions used in this paper are

$$F_{\mathrm{EH}} = \chi, \qquad F_{\mathrm{AX}} = \chi - \eta^2, \qquad F_{\mathrm{VT}} = \chi + \frac{q^2V^2}{9}, \tag{18}$$

$$F_{\mathrm{MX}} = \chi - \eta^2 + \frac{q^2V^2}{9} + \frac{2}{3}qV\eta\,\kappa^2\theta_{\mathrm{NY}}. \tag{19}$$

The auxiliary equations on AX, VT, and MX give $\eta = 0$, $V = 0$, and $(\eta, V) = (0, 0)$, respectively. On MX we retain the real-branch condition of Section 3.2. Therefore, for all modes and all topologies,

$$F_\Sigma\Big|_{\text{aux-shell}} = \chi \tag{20}$$

holds, and we obtain

$$\boxed{\dot{q}^2 + \chi = 0}. \tag{21}$$

This result is obtained independently of the diagnostic-level bridge result $C_{\mathrm{topology}} = -9\chi$ of Section 4.3, and constitutes the dynamical-side evidence of χ-universality (see Section 5.1).

7.2 χ-sign separation

Classifying the auxiliary-shell orbit equation by the sign of χ gives Table 9.

Sign of χ	Representative topology	Reduced vacuum behaviour	Notes
$\chi > 0$	S^3	$\dot{q}^2 = -\chi < 0$: no real initial data	A property of the vacuum orbit, independent of the Hamiltonian admissibility (Table 8)
$\chi = 0$	T^3	$\dot{q}^2 = 0$: static / degenerate	Stability discussion is outside the scope of this paper
$\chi < 0$	Nil^3, Sol^3	$\dot{q}^2 = -\chi > 0$: monotonic expansion and $q \to 0$ singular-approach collapse	Orbit classification within the vacuum reduced atlas of this paper

Table 9: χ-sign reduced orbit atlas.

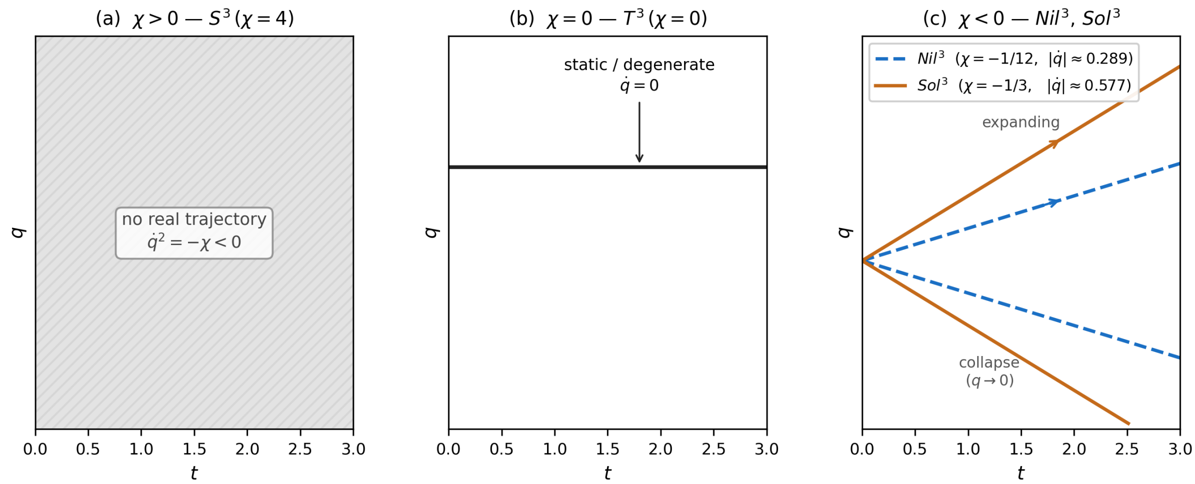

Figure 4: χ-sign reduced orbit atlas, schematic (q–t qualitative plots). **(a)** $\chi > 0$ (S^3, $\chi = 4$): no real initial data, since $\dot{q}^2 = -\chi < 0$ (hatched region). **(b)** $\chi = 0$ (T^3, $\chi = 0$): $\dot{q} = 0$, that is, static / degenerate. **(c)** $\chi < 0$ (Nil^3, Sol^3): two sheets $\dot{q} = \pm\sqrt{-\chi}$ (expanding / collapsing). The slope magnitudes are determined by $|\chi|$: Nil^3 gives $|\dot{q}| = \sqrt{1/12} \approx 0.289$ (dashed) and Sol^3 gives $|\dot{q}| = \sqrt{1/3} \approx 0.577$ (solid). Bounce-like / recollapse-like classes do not appear in any panel. *Caveats*: the figure is a qualitative schematic for the constant-velocity solutions of $\dot{q}^2 + \chi = 0$ and does not include an effective potential. Scope guard: matter coupling, anisotropic, or effective-source extensions can change the orbit classes (see Section 7.4 and Appendix C).

When $\chi > 0$ (S^3), the absence of real vacuum initial data is a property of the vacuum reduced atlas, and is independent of the admissibility classification of Table 8 (on which EH, AX, VT, and MX are all admissible). The introduction of matter coupling or an effective source can produce real orbits even on S^3, but such an analysis is outside the scope of the present paper.

7.3 Full-branch sweep summary

The classification of the reduced vacuum orbits across all 16 branches ($S^3/T^3/\mathrm{Nil}^3/\mathrm{Sol}^3$ $\times$ EH/AX/VT/MX) is as follows. Each $\chi < 0$ branch has two sheets (expanding and collapsing), so Table 10 counts 24 branch/sheet entries rather than 16 branches.

Orbit class	Count
`NO_REAL_AUX_SHELL_ORBIT`	4
`STATIC_OR_DEGENERATE`	4
`MONOTONIC_EXPANSION`	8
`SINGULAR_APPROACH`	8
`BOUNCE_LIKE`	0
`RECOLLAPSE_LIKE`	0

Table 10: Full-branch orbit class counts.

`NO_REAL_AUX_SHELL_ORBIT` reflects the absence of real initial data on S^3 ($\chi > 0$). `SINGULAR_APPROACH` is the orbit class on the collapse sheet of Nil^3 and Sol^3 ($\chi < 0$) for which $q \to 0$, and is not a statement about finite-time singularities or about observational cosmology. Neither class is the same question as the admissibility classification of Table 8.

7.4 Scope guard

The reduced vacuum atlas considered here contains no `BOUNCE_LIKE` or `RECOLLAPSE_LIKE` class. This is a statement about the vacuum isotropic reduced setting, and does not extend to matter-coupled, anisotropic, or effective-source models. The extensions needed to reach those settings are discussed in Section 8.3.

8 Discussion

8.1 What χ-universality organises

The results of this paper show that, in the Lorentzian EC+NY reduced sector, topology dependence is organised by the same scalar χ across three levels: Hamiltonian admissibility, P-channel diagnostics, and the reduced vacuum orbit.

At the admissibility level, the EH/AX/VT/MX mode pattern is common to the four topologies (Table 8), and the specific values of the topology do not alter the pattern itself. At the P-channel level, P_{form} is exactly zero independently of topology (Section 4.2) and Q_{NY} shows the mode separation (Section 4.4); neither has χ-dependence. Only the MX diagnostic P_{int} distinguishes the topologies, through $C_{\text{topology}} = -9\chi$ (Section 4.3). At the reduced-orbit level, the auxiliary-shell equation reduces to $\dot{q}^2 + \chi = 0$ (Section 7.1), and the qualitative classification of orbits is determined by the sign of χ (Section 7.2).

Taken together, the differences among the four topologies appear in the reduced sector as differences in the value of the scalar χ, while the remaining structure (the admissibility pattern, the behaviour of P_{form}, and the mode separation) is common across topologies. We call this phenomenon χ-controlled topology universality.

8.2 Relation to the paper IV response dictionary

The comparison with paper IV [5] suggests that the four Thurston geometries used throughout DPPU are not merely a convenient list of examples. In paper IV, these four geometries exhausted four response types under Euclidean KK Higgsing: none, uniaxial, triaxial, and biaxial. Here, in a Lorentzian spin-0 / isotropic EC+NY reduction, the same four geometries appear as structurally distinguished loci of the five-parameter structure-constant family: the origin (T^3), a single class-A axis (Nil^3), the isotropic class-A diagonal (S^3), and a balanced class-B pair (Sol^3). The χ-universality result therefore provides a new reading of the same four-geometry response dictionary in a different reduced sector, rather than a direct identification with the paper IV mechanism.

These two results suggest that the same set of Thurston geometries gives a coherent classificatory structure across different reduced sectors of DPPU theory, but the elucidation of a common underlying basis is left to future work.

8.3 Limitations and future directions

This paper is restricted to the $\alpha = 0$ EC+NY isotropic reduced sector, and the following extensions are explicitly left to future work.

For Nil^3 and Sol^3, this paper uses only the reduced description in terms of the isotropic scale q and does not derive the anisotropic EOM with independent scale factors. For

Sol^3, the construction of compact quotients, the classification of spin structures, and the determination of spectral data are outside the scope of the present paper. The absence of real initial data on the S^3 vacuum reduced atlas may be resolved by the introduction of matter coupling or an effective source, but that analysis is outside the scope of the present paper. As for the Weyl-source / $\alpha \neq 0$ branches, this paper is restricted to the $\alpha = 0$ baseline and does not include a re-evaluation of the other branches.

For the exact cancellation of P_{form} by block orthogonality and the χ-specialisation of $P_{\mathrm{int}}^{\mathrm{MX}}$, we present the results as confirmed by symbolic computation, and do not proceed to abstract formulation in the form of formal lemmas and propositions.

8.4 Closing remarks

In summary, this paper has shown that in the Lorentzian EC+NY reduced sector the four topologies S^3, T^3, Nil^3, Sol^3 share a common admissibility pattern, and that the topology dependence is collapsed onto a single scalar χ through two independent routes: the diagnostic identity $C_{\mathrm{topology}} = -9\chi$ and the reduced vacuum orbit $\dot{q}^2 + \chi = 0$. This χ-controlled topology universality is a classification result in the reduced sector; generalisation to anisotropic, matter-coupled, and global extensions is a future task beyond the scope of the present paper.

Acknowledgments

The author acknowledges the use of the following AI tools during manuscript preparation: Claude Opus 4.7 (Anthropic; accessed 2026-05), Gemini 3.1 Pro (Google; accessed 2026-05), Grok 4.3 (xAI; accessed 2026-05), and GPT-5.5 Thinking via ChatGPT (OpenAI; accessed 2026-05). These tools were used for language editing (including Japanese–English translation), outlining and improving exposition, drafting and reviewing auxiliary code (scripts and pseudocode) that was subsequently tested and validated by the author, and proposing candidate consistency checks and alternative derivations to be independently verified by the author.

The AI tools did not determine the scientific claims of this work and were not used to generate or modify research data or evidentiary figures. The author takes full responsibility for the content and for any remaining errors.

This work was developed within the informal collaborative project **DPPU** (**D**onut-like topology, **P**lanck-scale compactness, **P**recession dynamics, **U**niverse).

Availability and Contact

Code and Data Availability The code and supplemental data associated with this work are available at:

- Zenodo: 10.5281/zenodo.20228881
- GitHub: `https://github.com/Muacca/DPPUv7-paper06`

They are released under the MIT License.

Citation If you use these materials, please cite the associated manuscript and Zenodo record:

> Muacca, "Reduced-Sector χ-Universality in Lorentzian EC+NY Minisuperspace: Topology-Robust Admissibility, P-Channel Diagnostics, and Reduced Vacuum Orbits."

Contact For questions, bug reports, or collaboration inquiries:

- Email: `muacca@dmwp.jp`
- GitHub Issues: `https://github.com/Muacca/DPPUv7-paper06/issues`

A Topology conventions, local reductions, and caveats

A.1 Structure constants and χ values

The spatial structure constants for the four topologies are given as specialisations of the five-parameter LZ-native isotropic family. The family parameters are (a, b, c, u, v), and the antisymmetrised structure constants are defined by

$$C^1{}_{23} = \frac{a}{q}, \quad C^2{}_{31} = \frac{b}{q}, \quad C^3{}_{12} = \frac{c}{q}, \quad C^2{}_{12} = \frac{u}{q}, \quad C^3{}_{13} = \frac{v}{q}. \tag{22}$$

The three-dimensional Ricci scalar then reads

$$^{(3)}R\, q^2 = -\tfrac{1}{2}\,\Delta, \qquad \Delta = a^2 - 2ab - 2ac + b^2 - 2bc + c^2 + 4u^2 + 4uv + 4v^2, \tag{23}$$

and the χ of this paper is defined by

$$\chi = -\frac{\Delta}{12}. \tag{24}$$

The parameter values and χ values for each topology are listed in Table 11.

Topology	(a, b, c, u, v)	Δ	χ	Reduction status	Caveat
S^3	$(4, 4, 4, 0, 0)$	-48	4	closed isotropic reference	—
T^3	$(0, 0, 0, 0, 0)$	0	0	flat reference	—
Nil^3	$(0, 0, -1, 0, 0)$	1	$-1/12$	isotropic-scale local branch	anisotropic EOM left to future work
Sol^3	$(0, 0, 0, 1, -1)$	4	$-1/3$	isotropic-scale local branch	compact quotient outside the scope of this paper

Table 11: Topology conventions, χ values, and reduction status.

Verification. S^3: $\Delta = (16 - 32 - 32 + 16 - 32 + 16) = -48$, $\chi = 4$. Nil^3: $\Delta = 1$, $\chi = -1/12$. Sol^3: $\Delta = (4 - 4 + 4) = 4$, $\chi = -1/3$.

A.2 Distinguished loci in the five-parameter family

Table 12 records, on top of the topology ordering and the (Δ, χ) conventions fixed in Table 11, only the additional bridge-facing information. Here C_{topology} is computed from $C_{\text{topology}} = 3\Delta/4$ of Appendix B.1, and each row satisfies $C_{\text{topology}} + 9\chi = 0$.

Topology	$C_{\text{topology}} = 3\Delta/4$	Structural locus	paper IV echo
S^3	-36	isotropic class-A diagonal	triaxial
T^3	0	origin	none
Nil^3	$3/4$	single class-A axis	uniaxial
Sol^3	3	balanced class-B diagonal pair	biaxial

Table 12: Bridge-facing structural loci.

The paper IV echo column is interpretive only. It records a structural resonance with the response dictionary of paper IV, not an equivalence of mechanisms or a shared theorem.

A.3 Reduction scope for Nil^3 and Sol^3

Nil^3 is the non-Abelian homogeneous geometry determined by the left-invariant metric on the Heisenberg group [9, 10]; the sign convention adopted here is ${C^3}_{12} = -1/q$ (that is, $c = -1$). Under the isotropic-scale reduction a single scale parameter q is shared across all directions. The anisotropic EOM obtained by introducing independent scale factors is not derived in this paper.

Sol^3 is the geometry based on a solvable Lie group [9, 10] and enters the five-parameter family as a balanced class-B diagonal pair $u = 1$, $v = -1$. This is the isotropic-scale reduced entry of the present paper. A compact realisation of Sol^3 requires a mapping-torus construction; the choice of lattice normalisation, the classification of spin structures, and the determination of spectral data are outside the scope of the present paper. The Sol^3 entries of Table 8 and Table 9 are restricted to this local / isotropic reduced branch.

A.4 Lorentzian EC+NY action conventions

We adopt the orthonormal frame in Lorentzian signature $(-,+,+,+)$, and the Nieh–Yan coupling is parametrised by real θ_{NY}. The frame block-diagonal structure of P_{form} is common to the Lorentzian and Euclidean settings, but the sign convention of the Hamiltonian constraint follows the Lorentzian convention (Section 3.2). The definition of P_{int} and the orientation of Q_{NY} inherit the conventions of paper V [6].

The constant term $C_{\text{topology}} = -36$ of the S^3 specialisation agrees with the constant term of the MX-P expression of paper III ("Unified Geometric Landau EFT") [3]. On the other hand, the sign of the V^2q^2 term originates from the difference between the

Euclidean convention of paper III [3] and the Lorentzian metric-aware epsilon / legacy temporal-index map of the present paper.

B Computational checks and reproducibility

B.1 Algebraic core of the χ-bridge identity

We record the algebraic core of the family-level identity $C_{\text{topology}} + 9\chi = 0$ of Section 5.2. With the auxiliary quantity Δ defined as

$$\Delta = a^2 - 2ab - 2ac + b^2 - 2bc + c^2 + 4u^2 + 4uv + 4v^2, \tag{25}$$

the two quantities χ and C_{topology} are expressed as

$$\chi = -\frac{\Delta}{12}, \qquad C_{\text{topology}} = \frac{3\Delta}{4}. \tag{26}$$

Then

$$C_{\text{topology}} + 9\chi = \frac{3\Delta}{4} - \frac{9\Delta}{12} = 0. \tag{27}$$

This identity holds for arbitrary values of Δ and does not assume any specific topology. The specialised values for the four topologies (Table 11) are substitutions into this identity.

B.2 Full topology branch taxonomy

The main admissibility classification across all 16 branches ($S^3/T^3/\text{Nil}^3/\text{Sol}^3 \times$ EH/AX/VT/MX) is given in Table 13. Table 8 (Section 6) is the summary of this classification pattern. The full F-branch expressions, the Hamiltonian constraints, the auxiliary equations, and the auxiliary-shell solutions can be reproduced with `admissibility_classification.py`.

Topology	Mode	χ	F_{shell}	Hessian determinant	Label
S^3	EH	4	4	n/a	`L_ADMISSIBLE`
S^3	AX	4	4	-2	`L_ADMISSIBLE`
S^3	VT	4	4	$2q^2/9$	`L_ADMISSIBLE`
S^3	MX	4	4	$-4q^2(\kappa^4\theta_{\text{NY}}^2+1)/9$	`L_CONDITIONALLY_ADMISSIBLE`
T^3	EH	0	0	n/a	`L_ADMISSIBLE`
T^3	AX	0	0	-2	`L_ADMISSIBLE`
T^3	VT	0	0	$2q^2/9$	`L_ADMISSIBLE`
T^3	MX	0	0	$-4q^2(\kappa^4\theta_{\text{NY}}^2+1)/9$	`L_CONDITIONALLY_ADMISSIBLE`
Nil^3	EH	$-1/12$	$-1/12$	n/a	`L_ADMISSIBLE`
Nil^3	AX	$-1/12$	$-1/12$	-2	`L_ADMISSIBLE`
Nil^3	VT	$-1/12$	$-1/12$	$2q^2/9$	`L_ADMISSIBLE`
Nil^3	MX	$-1/12$	$-1/12$	$-4q^2(\kappa^4\theta_{\text{NY}}^2+1)/9$	`L_CONDITIONALLY_ADMISSIBLE`
Sol^3	EH	$-1/3$	$-1/3$	n/a	`L_ADMISSIBLE`
Sol^3	AX	$-1/3$	$-1/3$	-2	`L_ADMISSIBLE`
Sol^3	VT	$-1/3$	$-1/3$	$2q^2/9$	`L_ADMISSIBLE`
Sol^3	MX	$-1/3$	$-1/3$	$-4q^2(\kappa^4\theta_{\text{NY}}^2+1)/9$	`L_CONDITIONALLY_ADMISSIBLE`

Table 13: Compact full branch taxonomy.

B.3 Full orbit-sheet atlas

The 24 branch/sheet rows of the reduced vacuum orbit atlas are given in Table 14. S^3 and T^3 contribute one row per branch, while Nil^3 and Sol^3 contribute two sheets (expanding / collapsing). Table 10 (Section 7.3) summarises this branch/sheet count. The orbit equations, constraint residuals, and numerical monitors of each row can be reproduced with `orbit_atlas.py`.

Topology	Mode	Sheet	$\dot{q}_0$	Orbit class	Check
S^3	EH	none	no real solution	`NO_REAL_AUX_SHELL_ORBIT`	PASS
S^3	AX	none	no real solution	`NO_REAL_AUX_SHELL_ORBIT`	PASS
S^3	VT	none	no real solution	`NO_REAL_AUX_SHELL_ORBIT`	PASS
S^3	MX	none	no real solution	`NO_REAL_AUX_SHELL_ORBIT`	PASS
T^3	EH	zero	0	`STATIC_OR_DEGENERATE`	PASS
T^3	AX	zero	0	`STATIC_OR_DEGENERATE`	PASS
T^3	VT	zero	0	`STATIC_OR_DEGENERATE`	PASS
T^3	MX	zero	0	`STATIC_OR_DEGENERATE`	PASS
Nil^3	EH	expanding	$\sqrt{3}/6$	`MONOTONIC_EXPANSION`	PASS
Nil^3	EH	collapsing	$-\sqrt{3}/6$	`SINGULAR_APPROACH`	PASS
Nil^3	AX	expanding	$\sqrt{3}/6$	`MONOTONIC_EXPANSION`	PASS
Nil^3	AX	collapsing	$-\sqrt{3}/6$	`SINGULAR_APPROACH`	PASS
Nil^3	VT	expanding	$\sqrt{3}/6$	`MONOTONIC_EXPANSION`	PASS
Nil^3	VT	collapsing	$-\sqrt{3}/6$	`SINGULAR_APPROACH`	PASS
Nil^3	MX	expanding	$\sqrt{3}/6$	`MONOTONIC_EXPANSION`	PASS
Nil^3	MX	collapsing	$-\sqrt{3}/6$	`SINGULAR_APPROACH`	PASS
Sol^3	EH	expanding	$\sqrt{3}/3$	`MONOTONIC_EXPANSION`	PASS
Sol^3	EH	collapsing	$-\sqrt{3}/3$	`SINGULAR_APPROACH`	PASS
Sol^3	AX	expanding	$\sqrt{3}/3$	`MONOTONIC_EXPANSION`	PASS
Sol^3	AX	collapsing	$-\sqrt{3}/3$	`SINGULAR_APPROACH`	PASS
Sol^3	VT	expanding	$\sqrt{3}/3$	`MONOTONIC_EXPANSION`	PASS
Sol^3	VT	collapsing	$-\sqrt{3}/3$	`SINGULAR_APPROACH`	PASS
Sol^3	MX	expanding	$\sqrt{3}/3$	`MONOTONIC_EXPANSION`	PASS
Sol^3	MX	collapsing	$-\sqrt{3}/3$	`SINGULAR_APPROACH`	PASS

Table 14: Compact full orbit-sheet atlas.

B.4 Summary of main computational checks

Table 15 lists the logical verification targets. The first two checks are evaluated by the same symbolic script, because the family-level identity and its topology specialisations share the same derived (a, b, c, u, v) data.

Check	Content	Related section	Script
χ-bridge derivation	Extract (a, b, c, u, v) for $S^3/T^3/\mathrm{Nil}^3/\mathrm{Sol}^3$, derive χ from the LC spatial curvature, compute the raw $P_{\mathrm{int}}^{\mathrm{MX}}$ from the EC+NY MX curvature, and extract C_{topology}	Section 4.3, 5.2	`chi_bridge_symbolic.py`
P_{form} cancellation	Construct the curvature with the EC+NY engine for each torsion branch AX/VT/MX and compute P_{form}	Section 4.2	`pform_cancellation.py`
admissibility classification	Derive the lapse constraint and the auxiliary closure symbolically from the engine-derived F_{branch}	Section 3, 6	`admissibility_classification.py`
orbit atlas	Derive the orbit equation and the orbit class from the auxiliary shell of the engine-derived F_{branch}	Section 7	`orbit_atlas.py`

Table 15: Computational checks and their roles.

C Bianchi I lite comparison and reduced-atlas scope

C.1 Purpose

The purpose of this appendix is to make explicit the scope of the present paper by clarifying the structural differences between the reduced vacuum atlas of this paper and Bianchi I / Nieh–Yan-type systems.

C.2 Structural comparison

The reduced orbit equation of this paper is $\dot{q}^2 + \chi = 0$, which contains only the isotropic scale q and the topology scalar χ. It contains neither matter coupling nor effective-source terms. In contrast, the Bianchi I / NY-type bounce-like systems [12] typically involve several independent scale factors, matter sources, and effective potentials, and the existence of a turning point ($\dot{q} = 0$, $\ddot{q} > 0$) is a necessary condition for the realisation of a bounce-like orbit.

Item	Present reduced orbit atlas	Bianchi I / NY type
Orbit equation	$\dot{q}^2 + \chi = 0$	involves a richer effective potential
Scale factor	isotropic scalar q	generally several independent factors
Matter coupling	none	generally present
Bounce-like turning point	not found in the present scope	depends on the potential
Geometric basis	Thurston geometry $\times$ EC+NY torsion	model-dependent

Table 16: Structural comparison with Bianchi I lite.

This comparison is a sanity benchmark, not a matching of the same model.

C.3 Scope statement

The absence of `BOUNCE_LIKE` and `RECOLLAPSE_LIKE` orbit classes in the vacuum reduced atlas of the present paper is a property of this isotropic vacuum reduced setting. This result is not a general no-go claim against matter-coupled, effective-source, or anisotropic extensions. For instance, on S^3 ($\chi > 0$) the absence of real initial data on the vacuum reduced atlas is not a denial that the introduction of a matter source – changing the structure of the Hamiltonian constraint – could permit bounce-like orbits [12]. The analysis of these extensions is a future task.

References

[1] Muacca, "Topology-Dependent Phase Classification of Effective Potentials in Einstein–Cartan + Nieh–Yan Minisuperspace," Zenodo. `10.5281/zenodo.18213677` (2026).

[2] Muacca, "Structural Robustness of Isotropic S^3 Vacua in Einstein–Cartan Minisuperspace via Chiral Equilibrium and Weyl Stability," Zenodo. `10.5281/zenodo.18815498` (2026).

[3] Muacca, "Unified Geometric Landau EFT of Homogeneous $S^3 \times S^1$ Minisuperspace in Einstein–Cartan + Nieh–Yan Theory," Zenodo. `10.5281/zenodo.19144481` (2026).

[4] Muacca, "Homogeneous Three-Topology Comparison and Mode Dictionary in Einstein–Cartan + Nieh–Yan Theory: Geometric Structure of EC–Weyl Coupling," Zenodo. `10.5281/zenodo.19425147` (2026).

[5] Muacca, "Euclidean Geometric Response Dictionary and Selection Rules for Four Thurston Geometries," Zenodo. `10.5281/zenodo.19775695` (2026).

[6] Muacca, "Lorentzian Einstein–Cartan Minisuperspace with Nieh–Yan Torsion: Hamiltonian Branches, Pontryagin Diagnostics, and Weyl-Source Obstructions," Zenodo. `10.5281/zenodo.20084722` (2026).

[7] F. W. Hehl, P. von der Heyde, G. D. Kerlick, and J. M. Nester, "General relativity with spin and torsion: Foundations and prospects," Rev. Mod. Phys. **48**, 393–416 (1976).

[8] H. T. Nieh and M. L. Yan, "An identity in Riemann–Cartan geometry," J. Math. Phys. **23**, 373–374 (1982).

[9] W. P. Thurston, *Three-Dimensional Geometry and Topology, Volume 1*, edited by S. Levy (Princeton University Press, Princeton, NJ, 1997).

[10] P. Scott, "The geometries of 3-manifolds," Bull. London Math. Soc. **15**, 401–487 (1983).

[11] G. F. R. Ellis and M. A. H. MacCallum, "A class of homogeneous cosmological models," Commun. Math. Phys. **12**, 108–141 (1969).

[12] R. Brandenberger and P. Peter, "Bouncing Cosmologies: Progress and Problems," Found. Phys. **47**, 797–850 (2017), arXiv:1603.05834, doi:10.1007/s10701-016-0057-0.

[13] O. Chandia and J. Zanelli, "Topological invariants, instantons, and the chiral anomaly on spaces with torsion," Phys. Rev. D **55**, 7580–7585 (1997).